

Block 11a: shared board bootstrap in firmware unit tests

Date: 2026-09-28. Auditor: Codex /root. Source worktree: /private/tmp/kk715-11a. Immediate target: fork branch audit/715-11p-ci-prerequisites-20260928 at 0a6421e530fadf5bdc0b3b1160b962bbe5f7014d (PR #880), itself based on accepted Block 10 head 3a6f1cc2452c268908871958fe75abc5b3a2d54f (PR #876). Code generation predecessor: 781c75dc34357e4d828ca82c102f3e6f66264141; prerequisite merge: 23ae296ede1cdce7b2a42f9a25f00923c960275f. The containing report commit and hosted qualification belong in the PR body. Main-worktree master template SHA256 at original intake: 78d0fefefac9427057dc067cf4872d1240b53a7f5d68fa8c1cca18a1133d2f75.

1. Scope and reconciliation

This is the first bounded firmware-test unit after cumulative Block 11 PR #877 exhausted its authorized fourth review. A two-path manual-CI prerequisite unit 11p precedes it because the inherited Block 10 workflow needed exact scanner and report-test controls to qualify split branches. Three cumulative rounds required code changes. The fourth review 5345445076 on 2c53776877d751311850228c59277ab4e6b26ac9 found one confirmed test-harness root in comments 4127698063 and 4127698158. All five changed source paths here are assigned to 11a by the corrected split receipt. No firmware product source, protocol, build policy or dependency pin changes in this adjacent unit. Later units contain Solana runtime, storage, entropy, crypto, Windows and validator behavior.

The predecessor's passphrase fixture calls `timer_init()` directly if the layout canvas is absent. The test board helper already guards a single `kk_board_init()` per binary. Both functions initialize the same static runnable queue; a later board bootstrap can relink those nodes and make the periodic queue walk hang. The unit changes the fixture to call `kk_test_board_init()`, introduces one local declaration header, and uses it in two other changed test translation units. `confirm_test_utils.cpp` retains its existing declaration until its entire response-harness change lands in 11f. The board implementation itself is unchanged.

Direct and nested submodules were initialized at the predecessor pins and checked clean. Direct pins: `code-signing-keys` a6470bd8598e5e9a7bfc38bf139a5e5a616f05ec; `deps/crypto/trezor-firmware` 8a392f70a5d5575ece3dfb35f115d4a4b27f497c; `deps/device-protocol` 5fec9e6906a340be5eb3d795ec746769065b2db8; `deps/googletest` 7888184f28509dba839e3683409443e0b5bb8948; `deps/python-keepkey` 41af909341965c81b8afadd38f299d6f929089af; `deps/qrenc/QR-Code-generator` 6dfbfdad5d9303ed190d1c3cb7bec34b565b6ce8; `deps/sca-hardening/SecAESSTM32` 71d356a1141624994cf613bd2d2583892e8e6d5a.

2. Finding and security closure

The sensitive state is the static timer runnable queue in the emulator test process. The observer is any later confirmation or periodic test and the consequence is a blocked test process, rather than a device output. The entry path is `PassphraseTransition::SetUp()` before `AppConfirm` or other board users. Both full and Bitcoin-only native variants execute that fixture and the shared helper. The allowed behavior is one board initialization and completed test execution; the forbidden behavior is direct timer initialization followed by board initialization and a circular runnable queue. No alternate encoding or product-wire output applies to this harness state.

The source negative control at Block 10 base confirms the direct `timer_init()` call. The repaired source removes that call and invokes the guarded helper. An exploratory old-binary shuffled run stalled intermittently, while repeats also passed; it is not represented as a deterministic negative test. The fixed full and Bitcoin-only suites and passphrase-first shuffled tests are executed positive assertions. The two review comments share this root and are not counted as separate defects. The unrelated macro-recursion claim in comment 4127698115 was refuted on the cumulative PR because the macros are undefined before the wrapper definitions; reply 4127753705 and its resolved thread preserve that disposition.

Contract	Full	Bitcoin-only	Evidence
Native firmware suite	669 pass, 0 fail/error/skip	143 pass, 0 fail/error/skip	Raw XML and logs in archive
Passphrase then confirmation under Google Test shuffle seed 1	2 pass	2 pass	Shuffled logs in archive
Base source negative control	Direct timer call present	Same source	Manifest and predecessor source
Fixed source control	Guarded helper used; no direct timer call	Same source	Code predecessor and manifest

The seven-member archive 715-11a-controls-20260928.tgz has SHA256 db5cc31aea6d8d444ab0b075d9ab7116aae4e2577df534eelc01f2a98f8df0e5. Its manifest binds six raw files, the exact

base, code commit, outcomes and source control; a modified XML member fails the hash check. Full and Bitcoin-only binaries were built independently from the five-file unit checkout using the pinned Linux emulator toolchain. Hosted CI remains a separate exact-head gate; physical device execution is not claimed.

3. Local and external checkpoint

Local implementation and targeted/integration verification pass for the bounded harness defect. The source negative control distinguishes the old direct initialization from the fixed guarded path; the historical stall was intermittent and is not overstated. The pre-push `scripts/preflight.sh`, hosted CI, final report/PDF and exact PR head will be recorded after report freeze. No 11a Copilot review has been requested yet; its fresh budget is three rounds. The cumulative review verdict remains Changes recommended and is not relabeled as a pass. No release or physical-device acceptance is claimed.

4. Complete adjacent Git inventory

Adjacent inventory base: `0a6421e530fadf5bdc0b3b1160b962bbe5f7014d`. Original report predecessor: `38a42ea4946dbc81cd902e813c3cb7645bd09bf3`; prerequisite merge: `23ae296ede1cdce7b2a42f9a25f00923c960275f`. The final table includes this corrected report and PDF; reproduce from the immutable immediate predecessor to the final head recorded in the PR body with `git diff --numstat 0a6421e530fadf5bdc0b3b1160b962bbe5f7014d..FINAL_PR_HEAD`. Binary counts use Git's dash.

Added	Deleted	Path
-	-	docs/release/audit-units/715-11a-controls-20260928.tgz
45	0	docs/release/audit-units/715-11a-test-board-bootstrap-20260928.md
-	-	docs/release/audit-units/715-11a-test-board-bootstrap-20260928.pdf
1	2	unittests/firmware/app_confirm.cpp
2	7	unittests/firmware/storage_passphrase.cpp
4	3	unittests/firmware/test_board.cpp
4	0	unittests/firmware/test_board.h
2	2	unittests/firmware/usb_rx.cpp